



**AT
YOUR OWN
RISK**

① Find the least squares line $y = a + bx$ for the data.

x	-1	0	1	2	3	4	5	6
y	10	9	7	5	4	3	0	-1

$$y = a + bx$$

$$n\bar{y} = \sum y_i = na + b \sum x_i$$

$$\sum xy = a \sum x + \sum x^2 b$$

x	y	xy	x ²	n = 8
-1	10	10	1	
0	9	0	0	
1	7	7	1	
2	5	10	4	
3	4	12	9	
4	3	12	16	
5	0	0	25	
6	-1	-6	36	
<u>Σx = 20</u>	<u>Σy = 37</u>	<u>45</u>	<u>92</u>	

$$8a + 20b = 37$$

$$20a + 92b = 45$$

$$a = 7.45 \quad b = -1.13$$

$$y = 7.45 + (-1.13)x$$

② Fit a straight line $y = a_0 + a_1x$.

x	y	xy	x^2
1	1	1	1
3	2	6	9
4	4	16	16
6	4	24	36
8	5	40	64
9	7	63	81
11	8	88	121
14	9	126	196
56	40	364	524

NE

$$\sum y = na_0 + \sum x a_1$$

$$\sum xy = a_0 \sum x + a_1 \sum x^2$$

$$40 = 8a_0 + 56a_1$$

$$364 = 524a_0 + 56a_1$$

$$a_0 = 0.0958$$

$$b = 0.7279$$

③ Fit a straight line $y = ax + b$.

x	y	xy	x^2
6	5	30	36
7	5	35	49
7	4	28	49
8	5	40	64
8	4	32	64
8	3	24	64
9	4	36	81
9	3	27	81
10	3	30	100
72	36	479	569

NE

$$\sum y = a \sum x + nb$$

$$\sum xy = a \sum x^2 + b \sum x$$

$$36 = 722a + 96b$$

$$479 = 722a + 589b$$

$$y = 0.44x + 0.413$$

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$$y = a + bx + cx^2$$

NE =

$$\Sigma y = na + b \Sigma x + c \Sigma x^2$$

$$\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3$$

$$\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4$$

x	y	xy	x ²	x ² y	x ³	x ⁴
0	3.195	0	0	0	0	0
0.1	3.2299	0.32299	0.01	0.032299	0.001	0.0001
0.2	3.2532	0.65064	0.04	0.130128	0.008	0.0016
0.3	3.2611	0.97833	0.09	0.293499	0.027	0.0081
0.4	3.2516	1.30064	0.16	0.520256	0.064	0.0256
0.5	3.2282	1.61410	0.25	0.807050	0.125	0.0625
0.6	3.1807	1.90842	0.36	1.145052	0.216	0.1296
0.7	3.1266	2.18862	0.49	1.532037	0.343	0.2401
0.8	3.0594	2.44752	0.64	1.958016	0.512	0.4096
0.9	2.9759	2.67831	0.81	2.410479	0.729	0.6561
4.5	$\Sigma y = 31.7616$	$\Sigma xy = 14.0895$	$\Sigma x^2 = 2.85000$	$\Sigma x^2 y = 8.8288$	$\Sigma x^3 = 2.025$	$\Sigma x^4 = 1.533$

$$31.7616 = 10a + 4.5b + 2.85c$$

$$14.0895 = 4.5a + 2.85b + 2.025c$$

$$8.8288 = 2.85000a + 2.025b + 1.533c$$

The eqn is

$$y = 3.1746 + 0.5963x + (-0.9363)x^2$$

⑤. find a curve (parabola)
 $y = a + bx + cx^2$

$$\sum y = an + b \sum x + c \sum x^2$$

$$\sum xy = a \sum x + b \sum x^2 + c \sum x^3$$

$$\sum x^2 y = a \sum x^2 + b \sum x^3 + c \sum x^4$$

x	y	xy	x ²	x ² y	x ³	x ⁴
0	1	0	0	0	0	0
1	1.8	1.8	1	1.8	1	1
2	1.3	2.6	4	5.2	8	16
3	2.5	7.5	9	22.5	27	81
4	2.3	9.2	16	36.8	64	256

⑥ $\sum x = 10$ $\sum y = 8.9$ $\sum xy = 21.1$ $\sum x^2 = 30$ $\sum x^2 y = 66.3$ $\sum x^3 = 100$
 $\sum x^4 = 354$

$$8.9 = 5a + 10b + 30c$$

$$21.1 = 10a + 30b + 100c$$

$$66.3 = 30a + 100b + 354c$$

The eqⁿ is

$$y = 0.1302 + 1.95798x + (-0.3776)x^2$$

⑥. ~~fit a parabola of 2nd degree~~
 ~~$y = a +$~~

(6) Fit curve

$$y = ab^x \rightarrow y = ab^{x^2}$$

$$\log y = \log ab^x$$

$$\log y = \log a + x \log b$$

$$\log y = y$$

$$\log a = A$$

$$\log b = B$$

$$y = A + xB$$

$$2y = 2A + 2Bx$$

$$2xy = A2x + B2x^2$$

x	y	$y = \log y$	$2y$	$2x^2$
0	201	5.303	0	0
1	263	5.532	5.53	1
2	314	5.749	11.49	4
3	395	5.978	13.93	9
4	423	6.056	24.22	16
5	504	6.32	21.1	25
6	612	6.41	38.50	36
		41.299	120	140
28	2716			

$$41.299 = 2A + 28B$$

$$120 = 28A + 140B$$

$$A = 5.2135$$

$$B = 0.1215$$

$$a = 123.249$$

$$b = 1.187$$

Ans.

$$y = 123.249 \times 1.187^x$$

⑦

$$y = ae^{bx}$$

$$\log y = \log a + bx$$

$$\log y = A + bx$$

$$y = A + bx$$

$$\sum Y = nA + b \sum x$$

$$\sum xY = A \sum x + b \sum x^2$$

x	y	$Y = \log y$	$\sum xY$	$\sum x^2$	$\sum Y$
0	20	2.995	0	0	0
1	30	3.401	30	1	3.40
2	52	3.95	104	4	2.902
3	77	4.34	231	9	13.03
4	135	4.40	540	16	19.62
5	211	5.351	1055	25	26.75
6	326	5.78	1956	36	34.82
7	550	6.30	3850	49	44.16
8	1052	6.95	8416	64	55.66
			<u>16182</u>	<u>204</u>	<u>205</u>
<u>36</u>	<u>2453</u>	<u>44</u>			

$$44 = 9A + 36b$$

$$205, \underline{16182} = 36A + 204b$$

$$A = 2.955$$

$$b = 0.48$$

$$a = 19.2123$$

$$(0.48)x$$

$$y = (19.2123)e^{(0.48)x}$$

$$\log y = \log a + x \log b$$

⑧. $y = ab^x$

x	y	$y = \log y$	$x y$	x^2
2	144	4.969	9.93	4
3	172.8	5.1521	15.45	9
4	207.4	5.3346	21.33	16
5	248.8	5.5166	27.58	25
6	298.5	5.6987	34.192	36
	21	41	91	128

$$y = A + xB$$

$$\sum y = nA + \sum xB$$

$$\sum xy = A \sum x + B \sum x^2$$

$$91 = 5A + 21B$$

$$91 = 21A + 128B$$

$$A = 16.7689$$

$$B = -2.04$$

$$a = 19169752.4$$

$$b = 0.1300$$

$$y = (19169752.8)(0.130)^x$$

①

$$y = ax^b$$

$$\log y = \log a + b \log x$$

x	y	x	y	x^2	xy
61	350	4.110	5.85	16.89	24.0812
26	400	3.258	5.99	10.61	19.52
2	500	1.9459	6.21	3.78	12.09
2.6	600	0.9555	6.39	0.913	6.11
<u>96</u>	<u>1350</u>	<u>10.27</u>	<u>24.4</u>	<u>32.21</u>	<u>61.80</u>

$$y = A + bX$$

$$\sum y = nA + b \sum x$$

$$\sum xy = A \sum x + b \sum x^2$$

$$24.4 = 4A + 10.27b$$

$$61.8 = 10.27A + 32.21b$$

$$A = 6.4722 \quad b = -0.144$$

$$a = 646.905$$

$$y = (646.905)x^{(-0.144)}$$

find correlation

x	y	x^2	$x - \bar{x}$	y^2	$y - \bar{y}$	x^2	y^2	$x \cdot y$
600	2	360000	137.5	4	-20.5	18906.25	420.25	-2510
500	10	250000	37.5	100	-21.5	1406.25	462.25	-806.25
400	26	160000	-62.5	676	3.5	3906.25	12.25	-218.75
350	61	122500	-112.5	3721	38	12656.25	1482.25	-4311.25
<u>1850</u>	<u>99</u>	<u>900000</u>	<u>0</u>	<u>4501</u>	<u>0</u>	<u>36875</u>	<u>2377</u>	<u>-8175</u>

$$r = \frac{-8175}{\sqrt{36875 \times 2377}} = -0.873$$

④ correlatⁿ coef.

x	y	xy	x ²	y ²	xy
1	10	-10.2	1	100	10.19
2	12	-18.2	4	144	24.38
3	16	-24.2	9	256	48.37
4	28	-52.2	16	784	112.36
5	25	-50.2	25	625	125.35
6	36	-66.2	36	1296	198.34
7	41	-78.2	49	1681	296.33
8	49	-98.2	64	2401	392.32
9	40	-90.2	81	1600	360.31
10	50	-100.2	100	2500	400.30
	<u>307</u>	<u>0</u>	<u>550</u>	<u>19601</u>	<u>385.3</u>

7255

$$\frac{385.5}{\sqrt{82.5 \times 1962}} = 0.9581 \text{ u.}$$

(12) . calculate coef of correlation

x	y	X	Y	X ²	Y ²	XY
30	1	-16	-3	256	9	48
32	0	-14	-4	196	16	56
35	2	-11	-2	121	4	22
40	5	-6	1	36	1	-6
48	2	2	-2	4	4	-4
50	4	4	0	16	0	0
52	6	6	2	36	4	12
55	5	9	1	81	1	9
57	7	11	3	121	9	33
61	8	15	4	225	16	60
460	40	0	0	1092	64	230

$$r = \frac{230}{\sqrt{1092 \times 64}} = 0.875014$$

13.

IQ	IQ mark	Rank 10	Rank 9 exam	D = x - y	D ²
110	20	3	2	1	1
100	60	4	3.5	1.5	2.25
140	80	1	1	0	0
120	60	2	3.5	-1.5	2.25
80	10	6	6	0	0
90	20	5	5	0	0
					5.5

$$P_e = 1 - \frac{5.5 + \frac{2(4-1)}{12}}{6(36-1)}$$

$$= 1 - \frac{6}{35} = 0.828 \text{ //}$$

14.

X	Y	Rank X	Rank Y	D = x - y	D ²
10	30	10	10	0	0
15	42	6	3	3	9
12	45	9	2	7	49
12	46	4	1	3	9
13	33	8	9	-1	1
16	34	5	8	-3	9
24	40	2	4	-3	9
14	35	7	7	0	0
22	39	3	5	-3	9
20	38	3	6	-3	9
					104

$$S = \frac{1 - 6(1040)}{10(99)}$$

$$= \frac{0.369}{1} = 0.369$$

(15).

Fam	son	Rank _F	Rank _S	D _{F-S}	D _F
65	68	9	5.5	3.5	12.25
63	66	11	9.5	1.5	2.25
62	68	6.5	5.5	1	1
64	65	10	11.5	1.5	2.25
68	69	4.5	3	1.5	2.25
62	66	12	9.5	2.5	6.25
70	68	2	5.5	3.5	12.25
66	65	8	10	2.5	6.25
68	71	4.5	11	6.5	42.25
67	67	6.5	8	1.5	2.25
69	68	3	15.5	12.5	156.25
71	70	1.5	2	0.5	0.25
					22.5

30 x
68 → 2 times
62 → 2 times

30 y
68 → 2 times
65 → 2 times

$$\therefore \text{for } 1 - 6(22.5 + \frac{1(2(4-1))}{123})$$

$$= 1 - 6(22.5 + 2) / 12(12.5)$$

$$= 1 - \frac{6(24.5)}{12(12.5)} = 1 - \frac{6 \times 24.5}{12 \times 12.5}$$

$$= 0.239 //$$